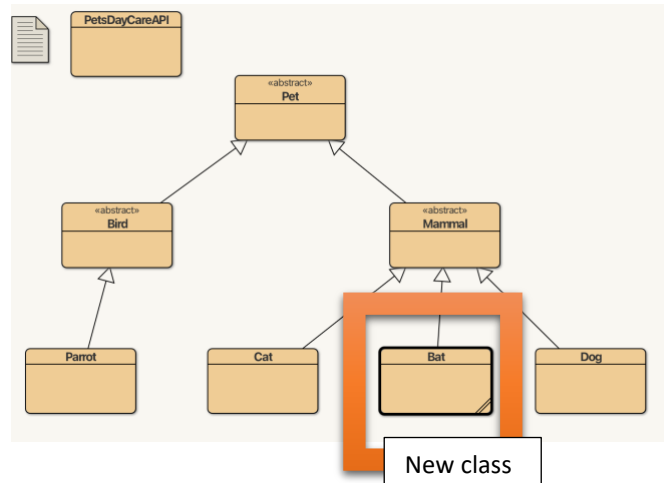


Class Diagram

All questions in this exam are based on one app.



Pet owner Owner id int age int name String daysAttending boolean[] setAge(int) void initName(String) void toString() String setId(int) void setName(String) void getName() String numOfDaysAttending() int setDaysAttending(boolean[]) void calculateWeeklyFee() double getId() int getDaysAttending() boolean[] checkIn(int) void getOwner() Owner checkOut(int) void getAge() int setOwner(Owner) void

You may detach this page from the booklet.

Name	Student Number	Course

Instructions:

- 2 hour exam.
- 4 Questions, answer all.
- Fill in your name, student number and course above.
- Complete the areas in this booklet and give it to your invigilator before leaving the room.
- Except for the front page, don't tear pages from this booklet; submit as a complete booklet.
- You can write anywhere in this booklet and use the back of pages for additional code, rough work, etc.

Question 1 - Models:

With this exam, you are given an inheritance hierarchy and the structure (UML) for the classes in the hierarchy. For the Pet, Mammal and Bat classes, code the following:

- The class definition
- The constructor
- The toString() method

Note: you do not have to define the fields or write the getters/setters.

Pet.java

Mammal.java

Bat.java

Question 2 - Controllers:

The class diagram has a PetDayCareAPI class. In this class, there is an ArrayList<Pet> defined:

```
public class PetDayCareAPI {  
  
    private ArrayList<Pet> pets;  
  
    public PetDayCareAPI () {  
        pets = new ArrayList<Pet>();  
    }  
}
```

Each box below has a comment in it. Code each method as defined in the comment.

PetDayCareAPI.java

```
// Write a public method, listAllPets(). The return type is String.  
// This method returns a list of the pets stored in the array list.  
// Each post should be on a new line and should be preceded by the  
// index number e.g.  
//      0: Pet 1 Details  
//      1: Pet 2 Details  
// If there are no posts stored in the array list, return a string  
// that contains "No Pets".
```

PetDayCareAPI.java

```
// Write a public method, numberOfDogs(). The return type  
// is int. This method returns the number of Dog objects  
// stored in the array list.  
// If there are none stored, zero is returned.
```

PetDayCareAPI.java

```
// Write a public method, listAllParrots().
// The return type is String.
//
// This method returns a list of the parrots that
// are stored in the array list
//
// Each matching post should be on a new line and should be preceded
// by the index number e.g.
//      1: Pet  Details
//      4: Pet Details
//
// If there are no pets stored in the array list, return a string
// that contains "No Pets".
//
// If there are no parrots in the list, the return string should
// have "No parrots here".
```

PetDayCareAPI.java

```
// If you included the sort functionality in your submitted code  
// base, please write the code to sort the array list by any criteria  
// you wish e.g. pet age.
```

Question 3 – Junit Tests:

This JUnit test class for PetsDayCareAPI contains test fixture data:

```
public class PetsDayCareAPITest {

    PetsDayCareAPI validAPI, emptyAPI;
    Owner owner_1;
    Dog doggie;
    Cat cattie;

    @BeforeEach
    void setUp() {
        file = new File("test.xml");

        validAPI = new PetsDayCareAPI("HappyPets", 30, "pets-1.xml");
        emptyAPI = new PetsDayCareAPI("NothingHere", 20, "pets-2.xml");

        owner_1 = new Owner(100, "Siobhan", "087-999");
        doggie = new Dog("Buddy", 12, owner_1, 12, 'M', true, 23, false, "BEAGLE", false);
        cattie = new Cat("kitty", 3, owner_1, 22, 'F', true, 5, true, true, "FEATHER WAND");

        validAPI.addPet(doggie);
        validAPI.addPet(cattie);
    }
}
```

Write the two test methods that are specified in the box below.

Note:

- You can assume that you don't have to write import statements.
- You can name the test methods anything you wish.

NewsFeedTest.java

In a previous question, you were asked to write a method called `numberOfDogs()`.

- This method returned the number of Dog objects stored in the array list.
- If there were none stored, zero is returned.

Using the test fixture data given above, write the two test methods, detailed below.

Test Method 1: This test method should verify that zero is returned by `numberOfDogs()` when no pets are stored in the `PetsDayCareAPI` object.

Test Method 2: This test method should verify that the correct number of dog objects stored is returned by `numberOfDogs()` when dog objects are stored in the `PetsDayCareAPI` object.

Question 4: Driver - runMenu()

In your assignment, you had a Driver class. If we take a simplified version of the menu you wrote, we just want the user to print out a message to the user each time an option is chosen. The usual structure for menu applies - loop until exit value is entered (use 0 as exit)

```
public class Driver {  
  
    public static void main(String[] args) {  
        new Driver();  
    }  
    public Driver() {  
        runMenu(); //TODO Write runMenu()  
    }  
}
```

This class displays the menu of options that the user can choose from - see the menu and a running of it below:

```
-----Simple Menu-----  
|  1) Print "hello"          |  
|  2) Print "well, now"     |  
|  3) Print "nothing to see here" |  
|-----|  
|  0) Exit                   |  
-----  
  
==>> 1  
hello  
  
-----Simple Menu-----  
|  1) Print "hello"          |  
|  2) Print "well, now"     |  
|  3) Print "nothing to see here" |  
|-----|  
|  0) Exit                   |  
-----  
  
==>> 0
```

Process finished with exit code 0

Driver.java

```
//add any methods required to show the user the menu  
//and loop until exit value is entered (use 0 as exit)
```